MOHAMMED SAHAL SHAIKH

Graduate Mechanical Engineer

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South African Permanent Resident | Code 10 driver's licence | Available immediately

PROFESSIONAL SUMMARY

Graduate mechanical engineer who designs, simulates and builds. BEng (Honours), Curtin University (Australia), Washington Accord-recognised and SAQA-evaluated at NQF level 8, with a 78% final-year average. Designed, built and instrumented a solar-panel test rig whose findings improved panel efficiency by 10.6%; completed GCAA-approved aircraft maintenance training; produced complete CAD and drawing packages in SolidWorks and Inventor; built an image-to-CFD thesis pipeline validated by physical testing. Seeking a graduate programme or graduate role in mechanical design, simulation, thermal or energy engineering in Gauteng.

Languages: English, Afrikaans and Urdu (fluent); Turkish (conversational).

EDUCATION

BEng (Honours), Mechanical Engineering | Curtin University (Australia), Dubai campus *2022 – 2026*

- Weighted average **72.89%**; final-year average **78%**; FEA and CFD units both above **80%**; Upper Second Class Honours.
- Washington Accord-recognised degree; **SAQA-evaluated at NQF level 8**.
- Honours thesis: cross-validated a porous-media model of eggplant tissue by image analysis (**4,498 pores**), ANSYS Fluent CFD (**within 0.17%**) and gravimetric testing (**96.6%** of theoretical uptake).

National Senior Certificate (Matric), Bachelor's pass | Ahmed Timol Secondary School, Azaadville *2019*

- Average **77%**, with a distinction in Physical Sciences (**80%**).

TECHNICAL SKILLS

CAD & Design: SolidWorks (incl. Simulation), Autodesk Inventor (incl. Frame Generator), AutoCAD, Onshape, FreeCAD; engineering drawings, GD&T, tolerance analysis, BOMs, DFM/DFA
Simulation & Analysis: ANSYS Fluent, ANSYS Mechanical, MATLAB, Simulink, SimScale; CFD, FEA, heat transfer, fatigue and Weibull reliability, FMEA, root cause analysis
Programming & Data: Python (OpenCV, pandas, scikit-learn), C++ (basic), Arduino (C/C++), Git, Advanced Excel
Energy & Sustainability: Solar PV testing, lifecycle assessment (openLCA), techno-economic analysis, Ashby material selection
Test & Manufacturing: Test-rig design and build, instrumentation and data acquisition, sheet metal, 3D printing, laser cutting, soldering
Project & Professional: MS Project, MS Office, project scheduling, risk registers, cost estimation, stakeholder management, technical report writing

PROFESSIONAL EXPERIENCE

Transnational Academic Group (operator of Curtin University Dubai, UAE): hired into three successive roles, 2023 – 2026; offered a permanent position.

Solar Energy Research Intern | Transnational Academic Group, Dubai *Feb 2026 – Jun 2026*

- Designed in SolidWorks, fabricated and instrumented a solar-panel soiling test rig, and developed the method used to quantify dust-driven losses in PV efficiency and service life.
- Ran controlled soiling tests with supporting CFD analysis; the resulting cleaning and maintenance schedule **improved panel efficiency by 10.6%**.
- Synthesised 20+ peer-reviewed studies of degradation mechanisms into technical reports.

Project Coordinator, ICISAS 2025 Conference | Transnational Academic Group, Dubai *Dec 2024 – Mar 2025*

- Led the peer-review operation: screened and triaged **450+ submissions** for scope and quality, then assigned each to reviewers from a **1,000+ reviewer** pool by technical expertise.
- Managed the coordination team and the end-to-end submission system to deliver reviews to the conference timeline.

Materials & Solar Research Intern | Transnational Academic Group, Dubai *Aug 2023 – Mar 2024*

- Compared **5 solar-panel technologies** on performance, degradation and end-of-life behaviour.
- Evaluated 2 aluminium-alloy combinations, analysing how selective laser melting affects microstructure and mechanical properties; co-authored the resulting **Springer Nature** publication.

STEM / CAD Instructor | The Logix Engine, Dubai *Oct 2025 – Feb 2026*

- Taught CAD and robotics to **~80 students across 6 schools** and coached a FIRST LEGO League team to the **national finals**.
- Designed and built hands-on student projects using soldering, wiring, 3D printing and laser cutting.

Aviation Maintenance Training Placement | Sarsan Aviation Academy (GCAA CAR-147), Dubai *Aug 2023*

- Completed supervised practical training on a Piper PA-28 across 26 aircraft components: sheet-metal repair, safety wiring, wing attachment and flight-control rigging, plus gas turbines, avionics and aviation regulations.

PROJECTS

- **Industrial gearbox design:** 1.6 kW, 8-year design life, <1% failure probability, with a full SolidWorks drawing package.
- **Heavy-duty standing desk redesign (2026, team of 5):** did most of the SolidWorks CAD and drawings (exploded views, BOM, detailed part drawings); House of Quality and Dominic concept selection targeting 40% less sway at full height.
- **Scale-model ATV (2021, team of 4):** did all of the Autodesk Inventor CAD: a 22-member tube chassis with Frame Generator, the full assembly and a complete drawing package.
- **Solar cooking backpack (James Dyson Award national finalist):** led the design of a portable solar cooker that cooked a meal in ~2 hours, 4 hours faster than traditional solar cookers.
- Also: residential renewable microgrid, sustainable footwear manufacturing process and a search-and-rescue robot. Full write-ups, drawings and the Honours thesis at msahal31100-dot.github.io/portfolio.

LEADERSHIP

- Founder of four student chapters at Curtin University Dubai: IEEE, ASHRAE, Robotics Club and the Federation of Economic Development Association (FEDA).
- President, IEEE Student Branch (2025 – 2026): led the inaugural ASPIRE SDG Innovation Challenge across 4 countries and co-led IEEE Convergence, including its first Simulation Engineering event.
- President, ASHRAE Student Branch (2025 – 2026): led the ASHRAE Poster Competition for three years; the 2025 edition drew 40+ entries from 12 UAE universities.
- Lead, PMI UAE Student Club (2025 – 2026) and Treasurer, Engineers Australia Student Chapter (2023 – 2026): owned event budgets, speakers and logistics, including a career-readiness event for 50–60 participants.

PUBLICATIONS

- Five peer-reviewed publications: Springer Nature (2025), selective laser melting of AlSi14 alloy; MARLOG 15 (2026), zero-emission logistics corridors; IEEE ASET 2025 (three papers). Full list on LinkedIn.

AWARDS

- 1st Place, Mubarmij50 Robotics Competition (2024), leading the Curtin Robotics Club team.
- 1st Place, Manipal University Dubai Hydrojet Competition (2022).
- 1st Runner-Up, Western Australia Government Climate Action Challenge (2024), with an official WA Government recognition letter.
- National Finalist, James Dyson Award (2023): solar cooking backpack (see Projects).
- Winner, ASHRAE Falcon Debate, Occupant Wellbeing team.

CERTIFICATIONS & TRAINING

- **Ansys (2026):** Supersonic, Subsonic and Two-Dimensional Flows; Shock-Expansion Theory; Flow over an Airfoil.
- **ZEISS Academy Metrology (2021):** GD&T Level 1 (Tolerancing of Size, Form, Location); GD&T Position; GD&T Profile Basics; Metrology Mathematics Level 1.
- **ASME:** GD&T Drawing Interpretation; Design for Sustainability; Industry 4.0 (two modules); Digital Literacy and Data Skills.
- **Royal Aeronautical Society & Vernewell Academy:** 5-week aerospace mentorship (space law, contracts, commercial space operations).

REFERENCE

Prof. Chithirai Pon Selvan, Head of School, Science and Engineering, Curtin University Dubai | Pon.selvan@curtindubai.ac.ae | +971 50 285 7651